

Milk intake and risk of hip fracture in men and women: a meta-analysis of prospective cohort studies.



Milk contains calcium, phosphorus, and protein and is fortified with vitamin D in the United States. All these ingredients may improve bone health. However, the potential benefit of milk on hip fracture prevention is not well established. The objective of this study was to assess the association of milk intake with risk of hip fracture based on a meta-analysis of cohort studies in middle-aged or older men and women. Data sources for this study were English and non-English publications via Medline (Ovid, PubMed) and EMBASE search up to June 2010, experts in the field, and reference lists. The idea was to compare prospective cohort studies on the same scale so that we could calculate the relative risk (RR) of hip fracture per glass of milk intake daily (approximately 300 mg calcium per glass of milk). Pooled analyses were based on random effects models. The data were extracted by two independent observers. The results show that in women (6 studies, 195,102 women, 3574 hip fractures), there was no overall association between total milk intake and hip fracture risk (pooled RR per glass of milk per day = 0.99; 95% confidence interval [CI] 0.96-1.02; Q-test $p = .37$). In men (3 studies, 75,149 men, 195 hip fractures), the pooled RR per daily glass of milk was 0.91 (95% CI 0.81-1.01). Our conclusion is that in our meta-analysis of cohort studies, there was no overall association between milk intake and hip fracture risk in women but that more data are needed in men. Copyright © 2011 American Society for Bone and Mineral Research.